

Secure University Attendance System

Capstone Project Proposal

Project Goal

Create a centralized mobile attendance platform that makes classroom attendance easier to manage while making proxy attendance significantly more difficult through multiple independent verification checks.

1. Problem Statement

University attendance is often recorded using different methods across courses, including paper sign-in sheets, online forms, quizzes, and manually maintained records. These approaches can be inconsistent, difficult to manage, and vulnerable to proxy attendance, where one student marks another student present. A centralized system could provide a consistent process while improving verification and record keeping.

2. Proposed Solution

The proposed system is a mobile attendance application for university classes. When a professor starts an attendance session, enrolled students receive a notification and press Attend. Before attendance is accepted, the system verifies the student's university identity, registered device, physical proximity, and identity confirmation. The design intentionally combines several checks rather than relying on only one method.

3. Core Verification Features

- **University Login:** Students sign in using their official university account, linking attendance to their real student identity and enrolled courses.
- **Device Binding:** Each student account is linked to one approved phone, making it more difficult to use a friend's phone or repeatedly switch devices.
- **Device Change Controls:** If a student replaces or loses a phone, a device-change request must be approved by an administrator or authorized university staff member.
- **Identity Confirmation:** The app can use the phone's built-in Face ID authentication where supported. As another option, app-based facial recognition could compare a live face with a university ID photo or a securely registered face profile.
- **GPS Presence Check:** The classroom or professor location defines an allowed attendance area. Students outside the configured radius cannot continue.
- **BLE Proximity Check:** The phone detects a temporary Bluetooth Low Energy signal from the professor's device or a classroom beacon, improving indoor proximity checking where GPS may be less accurate.

4. Attendance Process

1. Professor starts the session and enrolled students receive a notification.
2. Student signs in with the university account and presses Attend.
3. The app confirms the phone is the student's approved registered device.
4. GPS verifies that the student is within the allowed class or building radius.
5. BLE verifies that the phone can detect the active classroom signal.
6. The student completes the required Face ID or facial-recognition confirmation.
7. The backend validates the check-in; if it passes, the student is marked Present and the professor sees the result in real time.

5. Role-Based Dashboards

The application would provide a different dashboard for each type of user so that students, professors, and administrators only see the tools relevant to their role.

Student Dashboard	Shows active sessions, attendance history, absence count, ticket status, registered device information, and the option to request an approved device change.
Professor Dashboard	Allows professors to start or stop sessions, view live attendance, configure GPS/BLE settings, review tickets, manually correct attendance, and set course absence limits.
Administrator Dashboard	Handles device-change approvals, registered-device management, account or role administration, and review of audit logs or unusual verification activity.

6. Why the Multi-Factor Approach Is Stronger

The system does not depend on only one attendance method. A student would need to pass several independent checks at the same time, making common proxy-attendance methods much more difficult.

- Sharing a university password is not enough because attendance is also restricted to the approved device.
- Using a friend's phone is harder because the account is tied to a registered device.
- Students cannot repeatedly switch phones because device changes require approval.
- Being somewhere on campus is not enough because GPS can be combined with a classroom BLE proximity check.
- Having the correct phone is not enough if Face ID or facial recognition is also required.

7. Absence Tickets and Legitimate Exceptions

Students who miss class because of illness or another valid reason can submit an absence ticket through the application. The professor can review the request and decide whether the absence should remain absent or be changed to excused. Manual review or override can also be used when a legitimate device, accessibility, connectivity, or sensor issue prevents normal verification.

8. Scope and Evaluation

The first prototype would focus on university login, device registration, GPS/BLE presence checks, Face ID or facial-recognition confirmation, attendance recording, and the three role-based dashboards. Evaluation can measure check-in speed, classroom reliability, false accept/reject cases, BLE behavior in hallways and large lecture rooms, and overall usability for students and instructors.

9. Conclusion

This proposal combines university authentication, device binding, controlled device changes, identity confirmation, and GPS/BLE-based presence verification into a single attendance process. By requiring several independent checks, the system aims to make proxy attendance significantly more difficult while still giving students, professors, and administrators clear role-based tools for handling attendance and legitimate exceptions.

10. AI Assistance Disclosure

Artificial intelligence (AI) tools were used to assist with organizing, wording, and formatting this project proposal. The project concept, technical decisions, and final content were reviewed and developed by the student team.